2.1.10 Cluster

A cluster refers to the group of sectors that are manipulated as a unit. Depending on the filesystem and the size of the partition, the number of sectors that form a cluster changes. From the operating system's point of view, the cluster is the smallest storage unit.

2.1.11 Cylinders

A platter is made up of concentric tracks along which the read / write head stores data. The tracks are numbered. A cylinder refers to the same-numbered tracks of all the platters, since, theoretically, they form a cylinder.

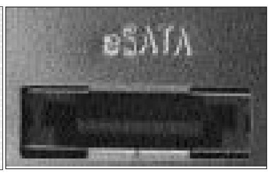
2.1.12 DMA/UDMA

Direct Memory Access / Ultra Direct Memory Access refers to a technology that allows a hard disk to manage data transfer without the aid of the CPU. This speeds up data transfers, while also leaving the CPU to perform other tasks. DMA / UDMA have gone through many iterations, the latest being UDMA 6, which offers transfer speeds of 133 Megabytes per second. DMA / UDMA modes are specified in the ATA standard. To achieve speeds above 33 Megabytes per second, as envisaged in UDMA3, a special 80-pin conductor cable is needed to reduce the interference that occurs between two data-carrying channels. Though the data is transferred using only 40 pins, the remaining pins are needed to ground interference created during the data transfer.

2.1.13 eSATA

External SATA is an interface that allows standalone SATA drives to be connected from outside the system. Since the SATA speeds extend to eSATA devices, eSATA offers the highest data transfer rates among all interfaces—namely, USB 2.0 and FireWire—at 3 Gigabits per second (in case of SATA 2).

An eSATA plug



The eSATA connector